

# Simple Probability

**Probability** is the likelihood, or chance, of something happening or not happening.

Probability is a ratio. It compares the number of favorable outcomes to the number of possible outcomes. A favorable outcome is the event you are analyzing.

probability of an event =  $\frac{\text{number of favorable outcomes}}{\text{total number of possible outcomes}}$

Probability is often written as a percent.

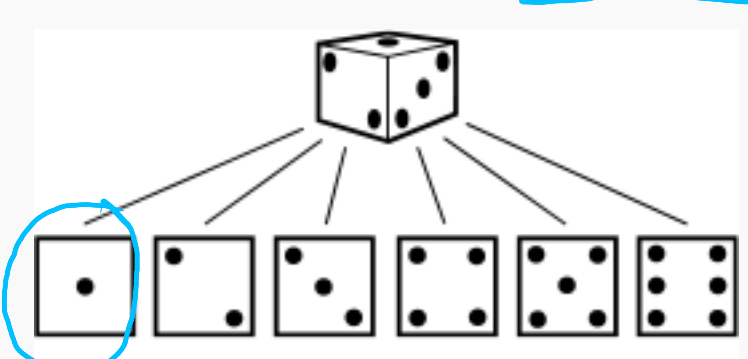
- A probability of 0% means than an event cannot happen.
- A probability of 100% means than an event is certain to happen.

Example 1 Hide

What is the probability of rolling a 6 using a regular 6-sided die?

Step 1

Count outcomes. Since the die has 6 sides, the total number of possible outcomes is 6. Since there is only 1 side with a 6, the number of favorable outcomes is 1.



Each face has an equal chance of turning up.

Step 2

Write the ratio.

probability of rolling a 6 =  $\frac{\text{number of favorable outcomes}}{\text{total number of possible outcomes}} = \frac{1}{6}$

The probability of rolling a 6 is  $\frac{1}{6}$  or **1 chance in 6** or about 17%.

You can also figure out the probability of something *not* happening.

Example 2 Hide

Oscar has 10 pairs of socks. Of the pairs, 3 are black, 2 are gray, 3 are brown, and 2 are blue. Without looking, Oscar reaches into the drawer and draws out a pair of socks. What is the chance that the socks are *not* blue?

The chance of getting blue is  $\frac{2}{10}$ , which equals  $\frac{1}{5}$ , or 20%.

To find the chance of *not* getting blue, subtract from 100%.  $100\% - 20\% = 80\%$

Finding the **odds** of an event happening is different from finding probability. To find the odds, write a ratio comparing the number of positive outcomes to the number of negative outcomes. The odds of Oscar drawing out a pair of blue socks are 2 blue to 8 non-blue, or 1 to 4. *odds*

Key Terms

 **probability**

Rate Your Confidence

^ High

= Medium

v Low

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Category

Lesson 5: Probability

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